

HARDWARE AND COMPONENTS BASICS

Day 4: GPUs and Cooling Systems

Theory: Basics of GPUs and Cooling Systems

A GPU (Graphics Processing Unit) is a computer component that processes images, videos, animations, and graphics. GPUs are useful for gaming, video editing, 3D designing, and other graphics-heavy tasks.

There are two main types of GPUs:

- Integrated GPU: Built into the processor and shares the computer's RAM.
- Dedicated GPU: A separate graphics card with its own memory (VRAM). It generally provides better performance for gaming and graphics work.

A computer also needs a cooling system to prevent components from becoming too hot. The two common types are air cooling and liquid cooling.

- Air cooling: Uses a heatsink and fan to remove heat.
- Liquid cooling: Uses a liquid coolant, pump, radiator, and fans to transfer heat away from the component.

Practical

Q1. Identify the GPU Model

GPU Brand: NVIDIA

GPU Model: GeForce RTX 3060

Type: Dedicated GPU

The NVIDIA GeForce RTX 3060 is a dedicated graphics card. It has its own VRAM and is designed for gaming, video editing, and other graphics-intensive applications.

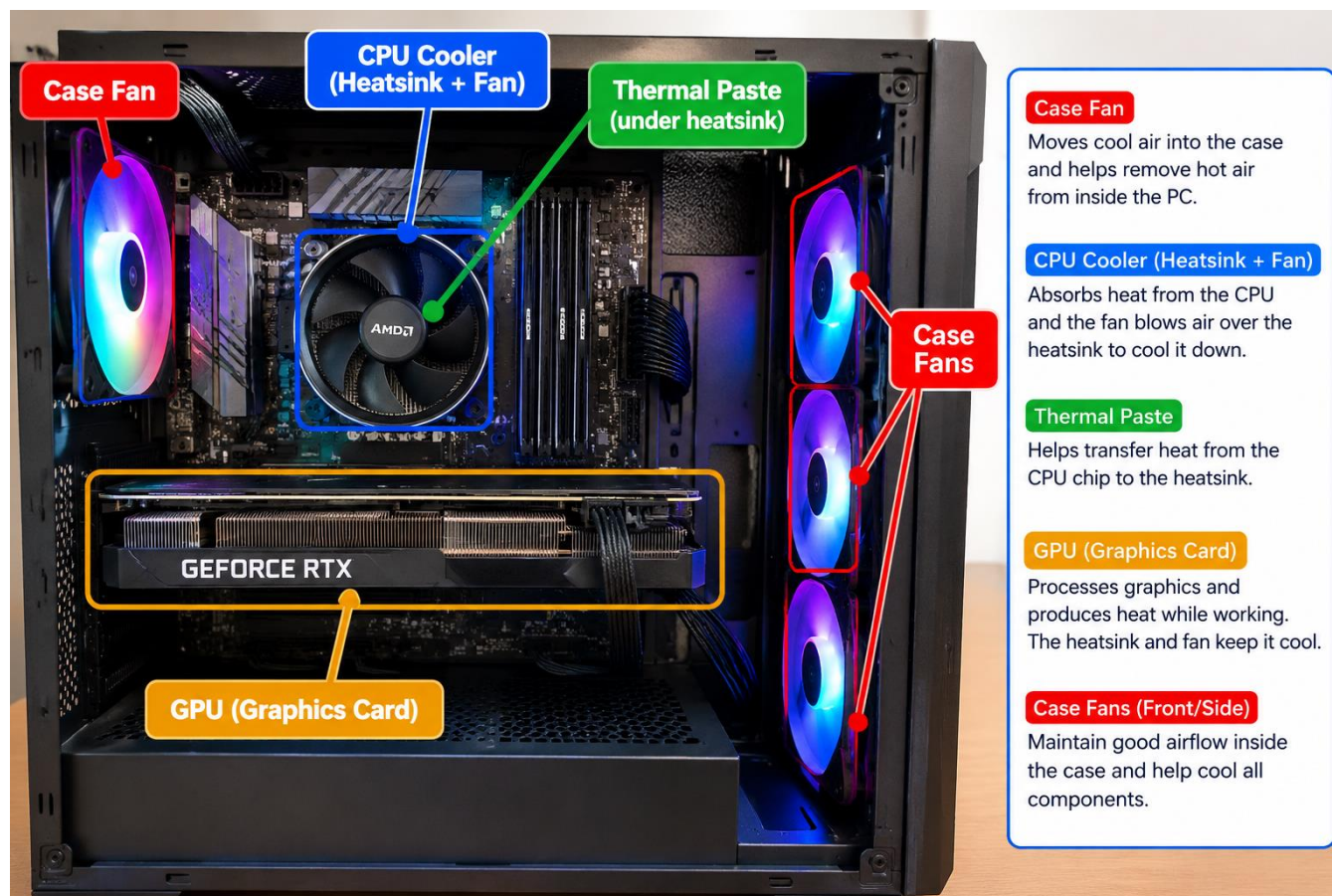
Q2. Cooling System

A typical gaming PC uses an air cooling system to keep the GPU and other components at a safe temperature.

Main Cooling Components:

- Fan – Moves air through the system and helps remove hot air.
- Heatsink – Absorbs heat from the GPU and spreads it over a larger surface area.
- Thermal Paste – Helps transfer heat from the GPU chip to the heatsink.
- Case Fans – Bring cool air into the PC and push hot air out.

Description: The GPU produces heat when it is working, especially while playing games. The heatsink absorbs this heat and the fan blows air over the heatsink to cool it down. Case fans also help maintain good airflow inside the computer.



Q3. Air Cooling vs. Liquid Cooling

Feature	Air Cooling	Liquid Cooling
Cost	Usually cheaper	Usually more expensive
Cooling Efficiency	Good for normal and gaming use	Generally better for high temperatures
Maintenance	Easy to maintain; mainly cleaning dust	Requires more care and checking for leaks
Installation	Simple installation	More complicated installation
Noise	Can become noisy at high speeds	Can be quieter in some setups
Space	Usually requires less planning	Requires space for radiator and pump

Conclusion: Air cooling is a simple and affordable option for most gaming computers. Liquid cooling can provide stronger cooling performance, but it usually costs more and is more complicated to install and maintain.

Q4. Choice for a Gaming and Streaming PC

If I were building a gaming PC for streaming IPL matches and playing graphics-heavy games, I would choose liquid cooling for the GPU system. Gaming and streaming at the same time can produce a lot of heat, so better cooling would help maintain stable performance. Liquid cooling can handle high temperatures effectively and can also help reduce thermal throttling. Although it costs more than air cooling, I think it would be a good choice for a powerful gaming and streaming PC.